



Name: Suggested Answers

CT: 26 Tutor:

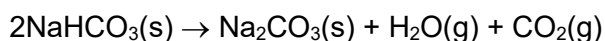
Marks

/ 8

To determine the percentage by mass of sodium hydrogencarbonate in a mixture

Sodium carbonate, Na_2CO_3 , does **not** decompose on heating with a Bunsen burner.

Sodium hydrogencarbonate, NaHCO_3 , decompose on heating.



- (a) You are to plan an experiment in which the percentage by mass of NaHCO_3 in a mixture of NaHCO_3 and Na_2CO_3 can be determined by heating and weighing alone.

You are to assume that only the following apparatus are available:

- boiling tube and holder,
- electronic mass balance,
- Bunsen burner,
- heat proof mat.

In your plan you should include details of:

- the procedure you would follow,
- the mass measurements you would make, including any other masses you would calculate from the experimental results (you may find it useful to label measurements in your plan as A, B, C etc.),
- how you would ensure that an accurate and reliable result is obtained.

Procedure

1. Using an electronic mass balance, weigh a clean and dry boiling tube.
2. Add the sample into the boiling tube and reweigh the boiling tube and its contents.
[1] for points 1, 2, 5
3. Heat the boiling tube and contents strongly for 10 minutes using the Bunsen burner.
4. Cool the boiling tube and contents on the heat proof mat.
[1] for points 3, 4
5. Re-weigh the boiling tube and contents.
6. Repeat steps 3 to 5 until constant mass readings for 2 consecutive weighing are obtained.
[1] for point 6

Results

Mass of empty boiling tube / g	A
Mass of boiling tube and sample / g	B
Mass of sample used / g	B – A
Mass of boiling tube and residue / g	C ₁ C ₂
Mass of residue / g	C ₂ – A

[1] correct headings and units

[1] raw data recorded as A, B, C, and

correct processed data for mass of sample and mass of residue / mass loss.

[5]

- (b) Outline how the labelled measurements in (a) can be used to determine the percentage by mass of NaHCO₃ in the mixture.

[A_r: H, 1.0; C, 12.0; O, 16.0; Na, 23.0]

Residue contains original Na₂CO₃ in mixture + Na₂CO₃ from decomposition of NaHCO₃.

Mass loss is due to CO₂ + H₂O.

$$\text{Mass loss} = B - C_2 \text{ g}$$

$$\text{Amount of CO}_2 \text{ and H}_2\text{O} = \frac{\text{mass loss}}{44.0+18.0} = \frac{B - C_2}{62.0} \text{ mol}$$

$$\text{Amount of NaHCO}_3 = 2 \times \frac{B - C_2}{62.0} = \frac{B - C_2}{31.0} \text{ mol}$$

$$\text{Mass of NaHCO}_3 \text{ in the mixture} = \frac{B - C_2}{31.0} \times 84.0 \text{ g [1]}$$

$$\text{Percentage by mass of NaHCO}_3 \text{ in the mixture} = \frac{\text{mass of NaHCO}_3}{B - A} \times 100 \% \text{ [1]}$$

[2]

- (c) By considering the products of the decomposition, suggest a reason why a crucible, without a lid, might be more appropriate than a boiling tube for this experiment.

- Water vapour from the decomposition might condense at the mouth of the boiling tube whereas the use of a crucible would enable the water vapour to escape more readily due to the wider opening.
- Hence, the use of a crucible will give more accurate mass readings. **OR**
Hence, the use of a crucible will lower the risk of breakage due to condensation.

[1] for both points

[1]